

# Topological Integrity Gravity: A Structural Horizon Transition

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## Abstract

We introduce Topological Integrity Gravity (TIG), a structurally constrained framework for gravitational horizon formation. The central idea is that admissible curvature corrections should not be treated as arbitrary phenomenological deformations of General Relativity, but as structures constrained by consistency requirements. Within a minimal static and spherically symmetric reduction, these conditions lead to a cubic normal form governing horizon formation. The resulting horizon equation exhibits a critical transition separating configurations with horizons from compact configurations without horizons. This transition is controlled by a dimensionless parameter and is characterized by the coalescence of roots in the associated cubic equation. We derive the critical parameter, analyze the discriminant structure, and discuss the interpretation of the transition as a structural threshold rather than a continuous perturbative deformation. The framework preserves the classical Schwarzschild limit while defining a constrained high-curvature sector suitable for further mathematical and observational analysis.

## 1 Introduction

General Relativity provides the most successful classical description of gravitation. Its geometric formulation identifies gravitational interaction with the curvature of spacetime and has passed a wide range of observational tests since Einstein's formulation of the gravitational field equations [1, 3, 4]. Nevertheless, the appearance of singularities and the structure of horizon formation indicate that the classical theory may not fully encode all structural constraints relevant at extreme curvature scales.

A common strategy is to modify the gravitational action by adding curvature corrections. Such models are often written as extensions of the Einstein-Hilbert action through a generalized function  $f(R)$  of the Ricci scalar. These approaches form a broad and well-studied class of modified gravitational theories [5–7]. However, this procedure may introduce an overly large space of possible theories. If curvature corrections are chosen freely, the resulting model space becomes phenomenological rather than structurally constrained.

The purpose of this work is to formulate a minimal alternative viewpoint. We assume that admissible curvature corrections are restricted by consistency conditions. In this sense, the question is not merely which correction terms can be written down, but which correction terms preserve the structural integrity of horizon formation.

We call this framework *Topological Integrity Gravity* (TIG). The term “topological integrity” is used here in a restricted mathematical sense: it refers to the preservation of the qualitative root structure associated with horizon formation under admissible curvature corrections.

The central result of this paper is the emergence of a cubic normal form,

$$x^3 - x^2 + \beta^3 = 0, \quad (1)$$

where  $x$  is a dimensionless radial horizon variable and  $\beta$  is a dimensionless structural control parameter. This equation exhibits a critical transition: for sufficiently small  $\beta$ , horizon solutions exist; beyond a critical value, no positive horizon root remains.

The transition is not introduced by hand. It follows from the algebraic structure of the reduced horizon equation. The resulting critical point is therefore interpreted as a structural boundary between horizon and non-horizon configurations. The associated root coalescence is structurally analogous to fold-type criticality and saddle-node bifurcations known from catastrophe theory [10].

## 2 Geometric and Physical Setup

We consider a four-dimensional spacetime  $(M, g)$  equipped with a Lorentzian metric  $g_{\mu\nu}$ . The discussion is restricted to static, spherically symmetric configurations. The general line element may be written in the form

$$ds^2 = -A(r)c^2dt^2 + B(r)dr^2 + r^2d\Omega^2, \quad (2)$$

where  $d\Omega^2$  denotes the metric on the unit sphere.

In the Schwarzschild solution of General Relativity, the horizon is located at the radial coordinate

$$r_s = \frac{2GM}{c^2}, \quad (3)$$

where  $M$  is the mass of the central object [2]. The Schwarzschild geometry remains the canonical reference solution for static black holes in General Relativity [3, 9].

The horizon condition can be expressed through the vanishing of the lapse function,

$$A(r) = 0. \quad (4)$$

In modified or structurally constrained theories, the functional form of  $A(r)$  may receive corrections. The key question is whether such corrections preserve a well-defined horizon structure.

**Definition 1** (Dimensionless horizon variable). *We define the dimensionless radial variable*

$$x := \frac{r}{r_s}. \quad (5)$$

*The Schwarzschild horizon corresponds to  $x = 1$ .*

**Definition 2** (Structural control parameter). *Let  $\ell$  denote a characteristic correction length scale associated with the regularization or structural modification of the central region. We define the dimensionless parameter*

$$\beta := \frac{\ell}{r_s}. \quad (6)$$

*The regime  $\beta \ll 1$  corresponds to corrections much smaller than the Schwarzschild scale.*

The limit  $\beta = 0$  must recover the classical Schwarzschild horizon. This requirement imposes a normalization condition on any admissible horizon equation: the dimensionless equation must admit  $x = 1$  as the physical root when  $\beta = 0$ .

### 3 Structural Consistency Principle

The guiding assumption of TIG is that gravitational corrections are not arbitrary deformations. Instead, they must preserve the qualitative integrity of the horizon formation problem.

**Definition 3** (Structural horizon consistency). *A horizon equation is structurally consistent if:*

- (i) it admits the Schwarzschild limit  $x = 1$  when the correction parameter vanishes;*
- (ii) it has a well-defined algebraic root structure;*
- (iii) horizon disappearance, if it occurs, takes place through a critical root coalescence rather than by arbitrary removal of solutions;*
- (iv) the correction terms are dimensionless and compatible with the chosen normalization;*
- (v) the resulting transition can be analyzed through invariant algebraic quantities such as the discriminant.*

These conditions are minimal. They do not specify a complete gravitational field theory. Instead, they constrain the admissible reduced horizon equations.

The simplest nontrivial algebraic setting in which a horizon can disappear through root coalescence is a cubic equation. A quadratic equation can describe a pairwise disappearance of roots, but does not naturally encode the Schwarzschild root together with a degenerate correction sector. The cubic normal form is therefore the minimal algebraic structure capable of encoding:

- (a) the Schwarzschild root at  $x = 1$ ;
- (b) a degenerate zero-radius sector;
- (c) a finite correction scale;
- (d) a discriminant-controlled structural transition.

This point is closely related to the normal-form viewpoint in local bifurcation analysis, where qualitative changes of solution structure are captured by minimal polynomial models [10].

## 4 Derivation of the Cubic Normal Form

We begin with the degenerate Schwarzschild-compatible polynomial

$$P_0(x) = x^2(x - 1). \quad (7)$$

This polynomial has a simple physical root at  $x = 1$  and a double root at  $x = 0$ . The root at  $x = 1$  represents the classical horizon, while the double root at the origin represents a degenerate internal sector.

Expanding (7), we obtain

$$P_0(x) = x^3 - x^2. \quad (8)$$

A finite structural correction must lift the degenerate zero-radius sector while preserving dimensional consistency. Since  $x$  is dimensionless and  $\beta$  is dimensionless, the lowest correction compatible with the cubic structure is proportional to  $\beta^3$ . We therefore define

$$P(x, \beta) = x^3 - x^2 + \beta^3. \quad (9)$$

The horizon equation is

$$P(x, \beta) = 0, \quad (10)$$

or explicitly,

$$x^3 - x^2 + \beta^3 = 0. \quad (11)$$

**Remark 1.** Equation (11) is not proposed as a fundamental field equation. It is a reduced algebraic normal form for the horizon condition under the structural consistency assumptions stated above.

## 5 Root Structure and Criticality

The cubic equation

$$x^3 - x^2 + \beta^3 = 0 \quad (12)$$

has a critical point when two real roots coalesce. This occurs when the polynomial and its derivative vanish simultaneously.

The derivative is

$$\frac{dP}{dx} = 3x^2 - 2x = x(3x - 2). \quad (13)$$

The nonzero stationary point is therefore

$$x_c = \frac{2}{3}. \quad (14)$$

At criticality, we impose

$$P(x_c, \beta_c) = 0. \quad (15)$$

Substituting  $x_c = 2/3$ , we obtain

$$\left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^2 + \beta_c^3 = 0. \quad (16)$$

This gives

$$\beta_c^3 = \frac{4}{27}, \quad (17)$$

and hence

$$\beta_c = \left(\frac{4}{27}\right)^{1/3}. \quad (18)$$

**Proposition 1** (Critical horizon transition). *The TIG horizon equation admits a critical transition at*

$$\beta_c = \left(\frac{4}{27}\right)^{1/3}. \quad (19)$$

*At this point, two real roots coalesce at*

$$x_c = \frac{2}{3}. \quad (20)$$

*Proof.* A double root occurs when  $P(x, \beta) = 0$  and  $P'(x, \beta) = 0$ . Since  $P'(x) = x(3x - 2)$ , the nonzero candidate is  $x = 2/3$ . Substitution into  $P(x, \beta) = 0$  yields  $\beta_c^3 = 4/27$ , completing the proof.  $\square$

## 6 Discriminant Analysis

The cubic equation

$$x^3 - x^2 + \beta^3 = 0 \quad (21)$$

can be written in the standard form

$$x^3 + ax^2 + bx + c = 0, \quad (22)$$

with

$$a = -1, \quad b = 0, \quad c = \beta^3. \quad (23)$$

The discriminant of a cubic is

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc. \quad (24)$$

Substituting  $a = -1$ ,  $b = 0$ , and  $c = \beta^3$ , we find

$$\Delta = 4\beta^3 - 27\beta^6. \quad (25)$$

Factoring,

$$\Delta = \beta^3(4 - 27\beta^3). \quad (26)$$

The discriminant vanishes when  $\beta = 0$  or when

$$\beta_c^3 = \frac{4}{27}. \quad (27)$$

The sign of the discriminant determines the real root structure:

- $\Delta > 0$ : three distinct real roots;
- $\Delta = 0$ : multiple roots;
- $\Delta < 0$ : one real root and two complex conjugate roots.

The discriminant therefore provides an invariant algebraic diagnostic of the structural transition.

## 7 Branch Geometry and Physical Interpretation

The positive roots of the horizon equation correspond to candidate horizon locations. In the Schwarzschild limit  $\beta = 0$ , the equation becomes

$$x^3 - x^2 = x^2(x - 1) = 0. \quad (28)$$

The physical nonzero root is

$$x = 1. \quad (29)$$

For small positive  $\beta$ , the physical horizon root is deformed away from the Schwarzschild value. As  $\beta$  increases, the relevant positive roots approach each other. At  $\beta = \beta_c$ , the roots coalesce at  $x = 2/3$ . For  $\beta > \beta_c$ , the corresponding pair of real roots no longer exists.

This is the structural horizon transition of TIG. It separates a regime in which horizon solutions exist from a regime in which compact configurations do not possess horizons. The transition is not interpreted as a smooth perturbation of the Schwarzschild radius. Instead, it represents a qualitative change in the root structure of the horizon equation.

## 8 Comparison with General Relativity

In General Relativity, the Schwarzschild horizon is located at  $r = r_s$ , or  $x = 1$  in dimensionless form. In TIG, the horizon condition becomes a cubic equation. The Schwarzschild solution is recovered in the limit

$$\beta \rightarrow 0. \quad (30)$$

Thus TIG does not remove the Schwarzschild solution from the theory. Instead, it embeds it into a larger structural setting in which the existence of horizons depends on the value of the control parameter  $\beta$ .

The classical horizon appears as the limiting member of a family of structurally constrained configurations. In this setting, General Relativity corresponds to a boundary point of the structural solution family rather than to an arbitrary member of a phenomenological parameter space.

## 9 Scaling Near Criticality

Near the critical point, root coalescence implies non-linear sensitivity. Let

$$\epsilon = \beta_c - \beta. \quad (31)$$

For small positive  $\epsilon$ , the separation between the coalescing roots is expected to scale as a fractional power of  $\epsilon$ .

To see this, expand  $P(x, \beta)$  near the critical point:

$$x = x_c + \delta, \quad \beta = \beta_c - \epsilon. \quad (32)$$

Since

$$P(x_c, \beta_c) = 0, \quad P_x(x_c, \beta_c) = 0, \quad (33)$$

the leading expansion is

$$P(x, \beta) \approx \frac{1}{2}P_{xx}(x_c, \beta_c)\delta^2 + P_\beta(x_c, \beta_c)(-\epsilon). \quad (34)$$

The derivatives are

$$P_{xx} = 6x - 2, \quad P_\beta = 3\beta^2. \quad (35)$$

At  $x_c = 2/3$ , one has  $P_{xx}(x_c) = 2$ . Hence

$$\delta^2 \sim 3\beta_c^2\epsilon, \quad (36)$$

and therefore

$$\delta \sim \epsilon^{1/2}. \quad (37)$$

This square-root scaling is characteristic of saddle-node type criticality [10]. Observable quantities depending on the horizon root may therefore exhibit nonlinear scaling near the transition.

## 10 Representative Effective Geometry

For illustration, one may consider a representative bounded-curvature lapse function of the form

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}, \quad (38)$$

where  $r_c$  is a regulated core scale. This expression is not introduced as a complete solution of new covariant field equations. It is a representative geometry used to examine the compatibility of the cubic horizon structure with a regularized high-curvature sector.

For large radial distance,  $r \gg r_c$ , one has

$$F(r) = 1 - \frac{2M}{r} + O\left(\frac{r_c^3}{r^4}\right), \quad (39)$$

so the Schwarzschild-compatible asymptotic behavior is recovered.

Near the origin, the lapse takes the form

$$F(r) = 1 - \frac{2M}{r_c^3}r^2 + O(r^5), \quad (40)$$

which corresponds to a regular de Sitter-like core in this representative sector. This does not constitute a general singularity-resolution theorem; it only states that the analyzed representative geometry has finite central behavior.

## 11 Photon Sphere Structure and Observable Deviation

A central consequence of the TIG framework is the modification of null geodesic structure near compact objects. In particular, the location of the photon sphere provides a direct probe of the underlying gravitational geometry.

The photon sphere is determined by the standard condition for unstable circular null geodesics:

$$rF'(r) - 2F(r) = 0. \quad (41)$$

Within the representative TIG geometry, the metric function takes the form

$$F(r) = 1 - \frac{2Mr^2}{r^3 + r_c^3}, \quad (42)$$

which introduces a characteristic scale  $r_c$  governing deviations from the classical Schwarzschild geometry.

In the limit  $r_c \rightarrow 0$ , the model smoothly reduces to the Schwarzschild solution with

$$r_{\text{ph}}^{\text{GR}} = 3M, \quad (43)$$

ensuring consistency with established gravitational theory.

We define the relative deviation from the Schwarzschild photon sphere as

$$\Delta_{\text{ph}} = \frac{r_{\text{ph}}^{\text{TIG}} - 3M}{3M}. \quad (44)$$

Numerical solutions of the representative model indicate a systematic inward shift of the photon sphere for finite core scale. The effect remains negligible for small  $r_c$ , but grows nonlinearly as the characteristic scale approaches the near-critical regime.

This shift implies a possible contraction of the photon sphere and hence a correction to the apparent shadow scale relative to General Relativity. At the present stage this should be interpreted as a model-dependent structural signature rather than a completed observational forecast.

## 12 Observable Implications

The TIG transition may affect observables sensitive to horizon-scale geometry. Possible implications include:

- modifications of black-hole shadow radius;
- altered photon-sphere structure;
- changes in near-horizon lensing behavior;
- modified ringdown or echo-like structures near criticality.

At the present stage, these implications should be regarded as structural predictions rather than fully developed observational forecasts. A complete analysis requires solving the full metric functions and perturbation equations. Nevertheless, the existence of a critical transition provides a clear target: observables should exhibit enhanced sensitivity near  $\beta = \beta_c$ .



## 13 Scope and Limitations

This paper develops the algebraic horizon structure of TIG. Several important questions remain open.

First, the complete dynamical field equations associated with the TIG consistency principle must be formulated. Second, stability of the resulting compact configurations must be analyzed. Third, observational predictions must be computed within a fully specified metric model.

The present work should therefore be read as the foundational normal-form analysis of the TIG horizon transition, not as a complete gravitational theory.

## 14 Conclusion

We have introduced Topological Integrity Gravity as a structurally constrained approach to horizon formation. Starting from minimal consistency requirements, we derived a cubic normal form for the horizon equation:

$$x^3 - x^2 + \beta^3 = 0. \quad (45)$$

This equation exhibits a critical transition at

$$\beta_c = \left(\frac{4}{27}\right)^{1/3}, \quad (46)$$

where two real roots coalesce at

$$x_c = \frac{2}{3}. \quad (47)$$

The resulting structure separates horizon and non-horizon configurations and suggests that gravitational systems may be constrained by algebraic consistency conditions beyond ordinary phenomenological deformation. Future work will focus on deriving full field equations, analyzing dynamical stability, and extracting observational signatures.

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